

V. EVALUATION

A. Evaluation Overview

This chapter evaluates the performance of the proposed Safe-Beam Assisted Power-Release Relay (SAPR-R) method, focusing on its ability to improve user satisfaction and service continuity under the aggregate EPFD constraint. The evaluation consists of three main parts. First, we observe the aggregate EPFD variation during the critical satellite pass over the GS, as well as the number of beams that must be shut down to satisfy the EPFD constraint. This analysis confirms the existence of EPFD violation and beam shutdown demand during the critical pass, and further supports the rationale of designing a relay-based recovery method. Second, we compare different interference mitigation methods in terms of service performance and relay capability. Finally, we further analyze the impact of different EPFD limits on user satisfaction to evaluate the stability and applicability of SAPR-R under different interference constraint conditions.

B. Simulation Setup

To evaluate the performance of the proposed SAPR-R method under the NGSO–GSO coexistence scenario, we design a simulation framework that combines STK and MATLAB. This framework is used to construct the LEO constellation, obtain the time-varying satellite geometry, and further evaluate the performance of different interference mitigation methods in terms of EPFD compliance, downlink capacity, and user satisfaction.

The simulation uses the following two main tools:

- **STK (Satellite Tool Kit):** [14] STK is used to construct the OneWeb-like LEO constellation and generate the relative positions, visibility relationships, and beam coverage information among the LEO satellites, the GS, and users at each time slot. Based on the system information generated by STK, the dynamic geometric variation during the critical satellite pass near the GS can be simulated.
- **MATLAB:** [15] MATLAB is used to process the system information generated by STK and perform offline preprocessing. This includes constructing the beam–user association, critical/helper satellite relationship, and critical-beam/helper-beam mappings. MATLAB is then used to execute the proposed SAPR-R method and calculate the EPFD and user satisfaction performance of each method at different time slots.

The main simulation parameters used in this chapter are summarized in Table I. The table integrates the settings related to the LEO constellation, GSO satellite, GS, LEO user ground station, and EPFD evaluation. These parameters are used in the following EPFD evaluation and user satisfaction comparison.

C. Compared Schemes

To evaluate the proposed SAPR-R method, we compare it with several interference mitigation methods. All methods are evaluated under the same simulation parameters to ensure a fair comparison. The compared schemes are described as follows:

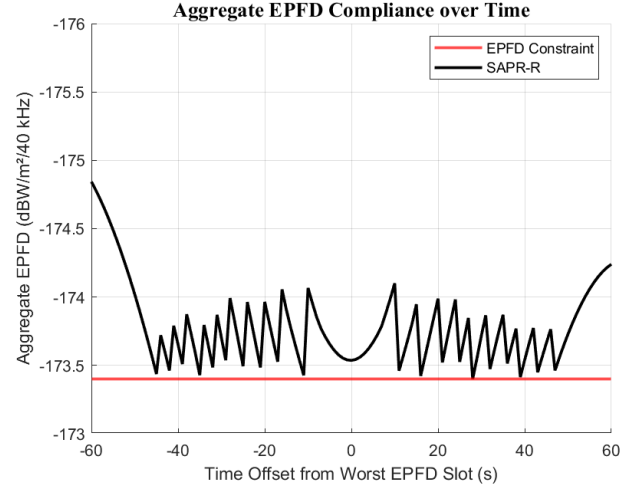


Fig. 1. Aggregate EPFD over time during the critical pass.

- **Beam shutdown only:** When the aggregate EPFD exceeds the limit, the high-interference beams on the critical satellite are directly shut down to satisfy the EPFD constraint. This method does not perform relay.
- **PC + Tilt:** This method represents a power-control and satellite-tilting based interference mitigation strategy. It reduces the interference toward the GS by adjusting the transmit power and satellite tilt angle. Although this method can help satisfy the EPFD constraint, it does not provide a relay-based recovery mechanism for users.
- **Only BPLR:** This method only applies the BPLR stage. Helper relay beams serve users located under the critical closed beams using the default power of the helper satellite. The SAHR stage is not applied.
- **SAPR-R (Proposed):** This method includes both the SAHR and BPLR stages.

D. Evaluation Results

Based on the simulation setup and compared schemes described above, this section evaluates the performance of SAPR-R during the critical pass. The analysis is divided into four parts. First, the aggregate EPFD and the number of closed beams are examined to verify whether SAPR-R can still satisfy the EPFD constraint after relay-based recovery, and to observe the beam shutdown behavior of the critical satellite at different time slots. Second, the average satisfaction of different methods is compared under different user loads to analyze how increasing system load affects service performance. Third, the number of relayed critical users is used to explain the main reason why SAPR-R improves satisfaction, and the satisfaction CDF is further used to observe the satisfaction distribution of all users. Finally, the average user satisfaction of SAPR-R under different EPFD limits is evaluated to observe how the strictness of the interference limit affects service performance.

1) EPFD Compliance and Beam Shutdown:

This subsection first analyzes the aggregate EPFD after applying SAPR-R and further observes the number of beams that must be shut down on the critical satellite to satisfy the

TABLE I
SIMULATION PARAMETERS.

Parameter	Value
LEO constellation	
Altitude (km)	1200
Orbit plane inclination	87.9°
Time slot	1 s
Maximum transmit power	16.8 W per satellite
Maximum antenna gain	34.33 dBi
Downlink frequency	11.7 GHz
Bandwidth	250 MHz
Antenna pattern	[12] $A \exp(\beta\phi)$
Maximum tilt angle	10°
Approximated antenna gain coefficient, (A)	4.0×10^3
Approximated antenna gain coefficient, (B)	-0.0671
GSO satellite	
Altitude (km)	35785.4
Longitude	30.6
Latitude	0
Downlink frequency	11.7 GHz
GSO ground station	
Minimum elevation angle	10°
Maximum receive gain	38.1 dBi
Noise temperature	240 K
Antenna pattern	[8]
LEO user ground station	
Maximum receive gain	40.95 dBi
Noise temperature	240 K
Antenna pattern	[8]
User demand	50 Mbps
EPFD	
EPFD limit (dB(W/m ² /1 MHz))	-173.4 [5]
Reference antenna diameter	60 cm

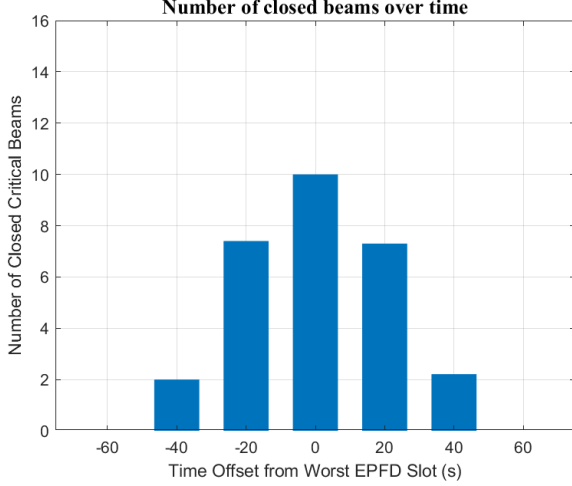


Fig. 2. Number of closed beams over time during the critical pass.

EPFD constraint. This analysis is used to verify whether the proposed method can still satisfy the EPFD limit after relay-based recovery, and to show the beam shutdown behavior during the critical pass.

Fig. 1 shows the aggregate EPFD variation during the critical satellite pass near the GS. The red line represents the EPFD constraint, and the black line represents the aggregate EPFD after applying SAPR-R. As shown in the figure, SAPR-R keeps the aggregate EPFD below the limit during the entire observation period. This indicates that the proposed

method does not cause EPFD violation while performing service recovery for closed-beam users. Therefore, SAPR-R can compensate for the service loss caused by beam shutdown while maintaining EPFD compliance.

Fig. 2 shows the number of beams that must be shut down on the critical satellite at different time offsets. When the satellite approaches the worst EPFD slot, the number of closed beams gradually increases and reaches its maximum around the worst EPFD slot. After the satellite moves away from GS, the number of closed beams gradually decreases. This result shows that the beam shutdown demand changes with the satellite-GS geometry.

2) User Satisfaction under Different User Loads:

This subsection compares the average satisfaction of different interference mitigation methods under different user loads. By changing the number of users, we can observe how the service performance changes as the system load increases, and evaluate whether SAPR-R can maintain user satisfaction under different load conditions.

Fig. 3 shows the average satisfaction under three user load scenarios: 30 users, 50 users, and 70 users. As shown in the figure, Beam shutdown only has the lowest satisfaction because it does not provide relay service for users affected by closed beams. PC + Tilt can reduce the interference toward the GS, but it does not recover the service of users affected by beam shutdown. Only BPLR can serve part of the closed-beam users through helper relay beams, so it achieves better satisfaction than Beam shutdown only and PC + Tilt.

Compared with the other methods, SAPR-R maintains higher average satisfaction under all three user loads. When

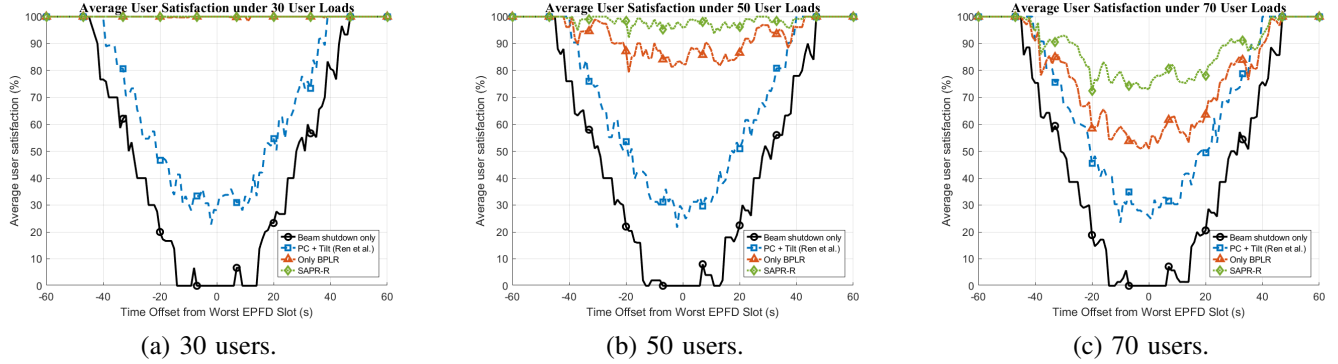


Fig. 3. Average satisfaction under different user loads. (a) 30 users. (b) 50 users. (c) 70 users.

the user load is low, the helper satellite usually has enough available power, so the difference between Only BPLR and SAPR-R is small. However, as the user load increases, more closed-beam users need relay service, and Only BPLR becomes limited by the default power of the helper satellite. In contrast, SAPR-R uses SAHR to release available power from the helper satellite and improve the relay capability of helper relay beams. Therefore, SAPR-R can maintain more stable service performance under medium and high user loads.

3) Relay Capability and Satisfaction Distribution:

This subsection further analyzes why SAPR-R improves user satisfaction and observes the satisfaction distribution of different methods. Since 30 users is a low-load scenario, the default available power of the helper satellite is usually sufficient to support most relay demand. Therefore, the difference in relay capability is not obvious under 30 users. To better show the effect of the proposed algorithm, this subsection focuses on the 50-user and 70-user scenarios.

Fig. 4 compares the number of successfully relayed critical users between Only BPLR and SAPR-R under the 50-user and 70-user scenarios. As shown in the figure, SAPR-R serves more critical users in both load scenarios. The difference becomes more obvious around the worst EPFD slot, where the beam shutdown demand is the most severe. This result shows the necessity of Stage 1 in SAPR-R.

Fig. 5 shows the satisfaction CDF of three interference mitigation methods. This figure focuses on the user-level satisfaction distribution. A CDF curve that shifts further to the right means that more users maintain higher satisfaction. In contrast, if the curve rises quickly in the low-satisfaction region, more users suffer from service degradation.

As shown in the figure, the CDF curve of SAPR-R shifts more toward the high-satisfaction region than the other methods. This means that more users can maintain better service quality. The result shows that SAPR-R not only improves average satisfaction, but also reduces the proportion of low-satisfaction users, making the overall user service quality more stable.

4) Impact of Different EPFD Limits:

This subsection further evaluates the impact of different EPFD limits on the service performance of SAPR-R. When the EPFD limit becomes stricter, the allowed aggregate interference becomes lower. As a result, the critical satellite may

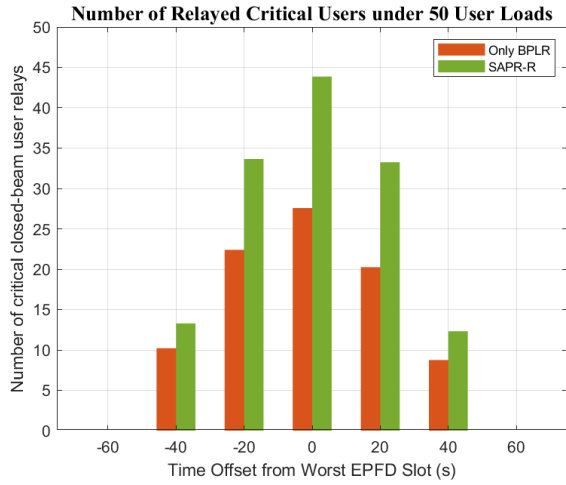
need to shut down more beams to satisfy the constraint, and the relay demand of closed-beam users may also increase. This analysis is used to observe whether SAPR-R can maintain stable average user satisfaction under different EPFD limits.

Fig. 6 shows the average user satisfaction of SAPR-R under different EPFD limits. As shown in the figure, when the EPFD limit is more relaxed, SAPR-R can maintain higher average user satisfaction. This result shows that the strictness of the EPFD limit directly affects user satisfaction. It also indicates that SAPR-R can achieve better service performance under more relaxed interference constraints.

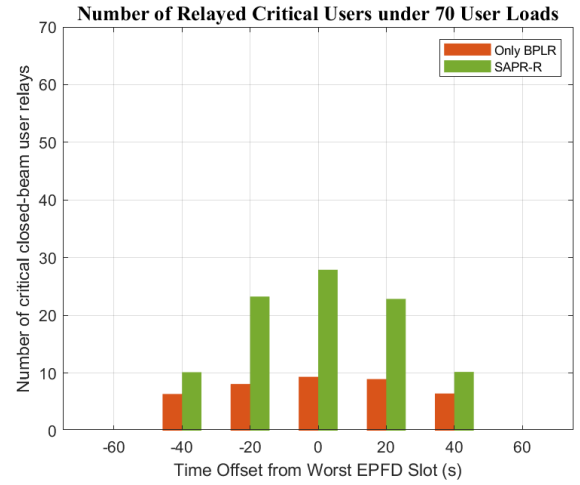
Overall, SAPR-R keeps the aggregate EPFD below the limit during the critical pass and uses SAHR to improve the ability of helper relay beams to serve critical users. Compared with other interference mitigation methods, SAPR-R maintains higher average satisfaction under different user loads and reduces the proportion of low-satisfaction users. In addition, the results under different EPFD limits show that the strictness of the interference constraint affects user satisfaction, and SAPR-R can maintain better service performance under more relaxed constraints. Overall, SAPR-R achieves a better balance between EPFD compliance and service continuity.

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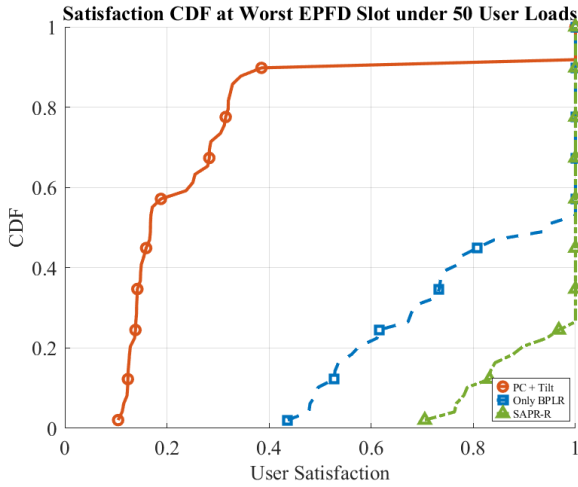


(a) 50 users.

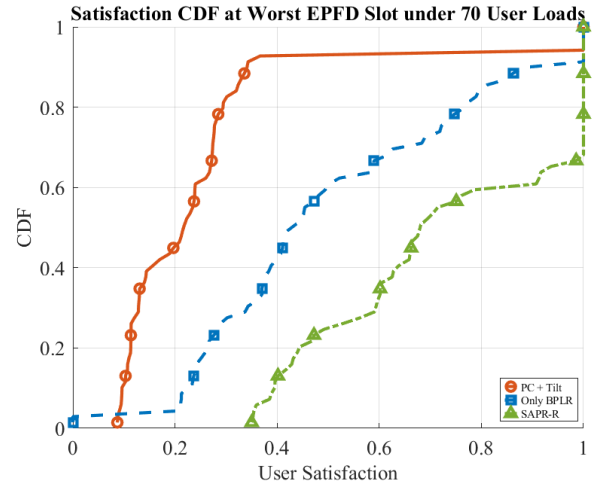


(b) 70 users.

Fig. 4. Number of relayed critical users with and without SAHR. (a) 50 users. (b) 70 users.



(a) 50 users.



(b) 70 users.

Fig. 5. CDF of user satisfaction under different interference mitigation methods.

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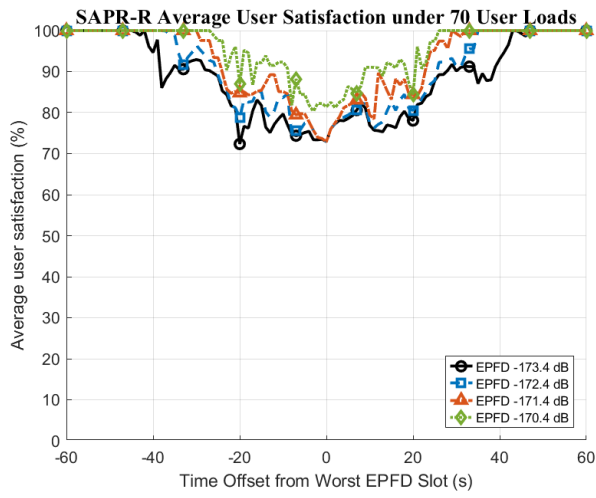


Fig. 6. Average user satisfaction of SAPR-R under different EPFD limits.